AFF–993

Sub. Code
4MMA1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

ALGEBRA – I

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define an Abelian group. Provide an example.

2. Define a normal subgroup. Give an example.

3. Define Normalizer of an element in G.

4. State third part of Sylow's theorem.

5. Define a commutative ring and give an example.

6. Define an integral domain.

7. Define an ideal of R with an example.

8. Define a maximal ideal of a ring with an example.

9. Define content of the polynomial.

10. When will you say a polynomial is integer monic?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) State and prove Lagrange's theorem.

Or

(b) If H is a nonempty finite subset of a group G and H is closed under multiplication, then prove that H is a subgroup of G.

12. (a) If G is a finite group then prove that $C_a = o(G) | o(N(a))$.

Or

(b) If P is a prime number and $P | o(G)$, then prove that G has an element of order P.

13. (a) Prove that a finite integral domain is a field.

Or

(b) If R is a ring, then prove that $\forall a, b \in R$.

(i) $a0 = 0a = 0$

(ii) $a(-b) = (-a)b = -(ab)$

(iii) $(-a)(-b) = ab$.

14. (a) Let R, R' be rings and ϕ a homomorphism of R onto R' with Kernel U. Then prove that R' is isomorphic of $R|U$.

Or

(b) Prove that every integral domain can be imbedded in a filed.

15. (a) Let R be a Euclidean ring. Suppose that for $a,b,C \in R$, $a|bc$ but $(a,b)=1$. Then prove that $a|c$.

Or

(b) If p is a prime number of the form $4n+1$, then prove that $p=a^2+b^2$ for some integers a,b.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Let ϕ be a homomorphism of G onto $\overline{G}$ with Kernel K. Then prove that $G|K \cong \overline{G}$.

17. If p is a prime number and $p^{\alpha}|o(G)$, then prove that G has a subgroup of order p^{α}.

18. If ϕ is a homomorphism of R into R' with Kernel $I(\phi)$, then prove that

(a) $I(\phi)$ is a subgroup of R under addition.

(b) If $a \in I(\phi)$ and $r \in R$ then both ar and ra are in $I(\phi)$.

19. If R is a commutative ring with unit element and M is an ideal of R, then prove that M is a maximal ideal of R if and only if R/M is a field.

20. Prove that $J[i]$ is a Euclidean ring.

AFF–994

Sub. Code
4MMA1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

ANALYSIS – I

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. If p is a limit point of a set E then prove that every neighborhood of p contains infinitely many points of E.

2. If F is closed and K is compact then prove that $F \cap K$ is compact.

3. If $0 \le x < 1$, then $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$. If $x \ge 1$ then prove that the series diverges.

4. Test the convergency of the series $\sum_{1}^{\infty} 1/n^3$.

5. Given the power series $\sum C_n Z^n$, put $\alpha = \lim_{n\to\infty} \sup \sqrt[n]{|C_n|}$, $R = 1/\alpha$. Then prove that $\sum C_n Z^n$ converges if $|Z| < R$ and diverges if $|Z| > R$.

6. Find the radius of convergence of $\sum n^3 z^n$.

7. Prove that monotonic functions have no discontinuities of the second kind.

8. Let f be a continuous real function on the internal $[a, b]$. If $f(a) < f(b)$ and if C is a number such that $f(a) < c < f(b)$ then prove that there exists a point $x \in (a,b)$ such that $f(x) = c$.

9. If f and g are continuous real functions on $[a, b]$ which are differentiable in (a, b), then prove that there is a point $x \in (a, b)$ at which

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x).$$

10. Let f be defined for all real x and suppose that $|f(x) - f(y)| \le (x - y)^2$ for all real x and y. Prove that f is constant.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Prove that every K-cell is compact.

Or

(b) Let P be a non-empty perfect set in R^k. Then prove that P is uncountable.

12. (a) Suppose $\{S_n\}$ is monotonic. Then prove that $\{S_n\}$ converges if and only if it is bounded.

Or

(b) If $P > 1$, $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ converges. If $P \le 1$ the series diverges. Prove.

13. (a) Suppose (i) $|C_1| \geq |C_2| \geq |C_3| \geq \cdots$ (ii) $C_{2m-1} \geq 0,\ C_{2m} \leq 0$ $m = (1, 2, 3\cdots)$ (iii) $\lim_{n\to\infty} C_n = 0$. Then prove that $\sum C_n$ converges.

Or

(b) Prove that the convergence of $\sum a_n$ implies the convergence of $\sum \frac{\sqrt{an}}{n}$ if $a_n \geq 0$.

14. (a) Prove that a mapping f of a metric space X into a metric space Y is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set V in Y.

Or

(b) Let f be monotonic on (a, b). Then prove that the set of points of (a, b) at which f is discontinuous is atmost countable.

15. (a) Let f be defined on $[a, b]$; if f has a local maximum at a point $x \in (a, b)$ and if $f'(x)$ exists then prove that $f'(x) = 0$.

Or

(b) Suppose f is a real differentiable function on $[a, b]$ and suppose $f'(a) < \lambda < f'(b)$. Then prove that there is a point $x \in (a,b)$ such that $f'(x) = \lambda$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If a set E in R^k has one of the following three properties then prove that it has the other two

(a) E is closed and bounded

(b) E is compact

(c) Every infinite subset of E has a limit point in E.

17. (a) If $\overline{E}$ is the closure of a set E in a metric space X then prove that diam $\overline{E}$ = diam E.

(b) If K_n is a sequence of compact sets in X such that $K_n \supset K_{n+1}$ $(n = 1, 2, 3, \cdots)$ and if $\underset{n\to\infty}{Lt}$ diam $K_n = 0$ then prove that $\bigcap_{1}^{\infty} K_n$ consists of exactly one point.

18. Suppose (a) $\sum_{n=0}^{\infty} a_n$ converges absolutely (b) $\sum_{n=0}^{\infty} a_n = A$ (c) $\sum_{n=0}^{\infty} b_n = B$ (d) $C_n = \sum_{K=0}^{n} a_k b_{n-k} (n = 0, 1, 2 \cdots)$ Then prove that $\sum_{n=0}^{\infty} C_n = AB$. That is the product of two convergent series converges, and to the right value, if atleast one of the two series converges absolutely.

19. If f and g be complex continuous functions on a metric space X. Then prove that $f + g$, fg and f/g are continuous on X. In the last case assume $g(x) \neq 0$, for all $x \in X$.

20. State and Prove Taylor's theorem.

AFF–995

Sub. Code
4MMA1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

DIFFERENTIAL GEOMETRY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write down Seret-Frenet formulae.

2. Define the osculating plane at a pt of a space curve.

3. Find the equation of osculating sphere at $(1,2,3)$ on the curve $x = 2t+1 \quad y = 3t^2+2 \quad z = 4t^3+3$.

4. Define involute and evolute of a curve.

5. Is $x = a\cos t \quad y = a\sin t \quad z = ct$ a plane curve.

6. Find the equ of tangent plane for $z = x^2 + y^2$ at $(1,-1,2)$.

7. Calculate fundamental magnitudes for surface of revolution $x = u\cos v$, $y = u\sin v$, $z = fu)$.

8. State MeuSNIER's theorem.

9. Write down the differential equation of lines of curvature.

10. State Monge's theorem.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Show that the Serret-Frenet formulae can be written in the form $\bar{t}' = \bar{w} \times \bar{t}$, $\bar{n}' = \bar{w} \times \bar{n}$, $\bar{b}' = \bar{w} \times \bar{b}$.

Or

(b) Calculate curvature and torsion of the curve $r = \left[a\left(3u - u^3\right), 3au^2, a\left(3u + u^3\right)\right]$.

12. (a) Show that the locus of centre of curvature is an evolute only when the curve is plane.

Or

(b) Show that $R^2 = \left| \frac{\bar{t} \times \bar{t}''}{k^2 T} \right|^2$ where R is the radius of spherical curvature.

13. (a) Derive the fundamental magnitude and normal to the surface $\alpha Z = ax^2 + 2hxy + hy^2$.

Or

(b) Show that the metric is positive quadratic form in du , dv .

14. (a) Find the differential equation of geodesic via normal property.

Or

(b) Find the equ of geodesic on the merge's form of the surface $z = f(x, y)$.

15. (a) State and prove Merge's theorem.

Or

(b) Prove that $xy = (z - c)^2$ is developable.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Calculate the curvature and torsion of cubic curve $\bar{r} = (u, u^2, u^3)$.

    (b) Find radii curvature of the helix $x = a\cos u$, $y = a\sin u$, $z = au\tan\alpha$.

17. State and prove uniqueness theorem for space curves.

18. State and prove Existence theorem for space curves.

19. State and prove Bonnet theorem.

20. (a) Find the line of curvature of parabolid $xy = az$.

    (b) Find the differential equation of line of curvature of helicoid $x = u\cos v$ $y = u\sin v$, $z = f(u) + cv$ and hence find the condition for the meridians to be lines of curvature.

AFF–996

Sub. Code
4MMA1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

DIFFERENTIAL EQUATIONS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. State Uniqueness theorem.

2. Write the formula to compute particular solution of $L(y) = b(x)$.

3. Find the regular singular points of $(1 - x^2)y'' - 2xy' + 2y = 0$.

4. Define indicial polynomial.

5. Eliminate the function f from $z = x + y + f(xy)$.

6. Define characteristic ship.

7. Write the one dimensional diffusion equation.

8. When $F(D, D')$ is reducible?

9. State interior Dirich let problem.

10. State the interior Churchill problem.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Prove that there exist n linearly independent solutions of $L(y)=0$.

Or

(b) Verify $\phi_1(x)=x, x>0$ satisfies the given equation and find a second independent solution, $y''-2xy'+2y=0$.

12. (a) Find all solutions of $x^2y''+xy'-4y=x$.

Or

(b) Obtain two linearly independent solution of $x^2y''+2x^2y'-2y=0$.

13. (a) Find the general integral of $y^2p-xyq=x(z-2y)$.

Or

(b) By Charpits method find the complete integral of $p^2x+q^2y=z$.

14. (a) Solve : $r+s-2t=e^{x+y}$.

Or

(b) Reduce the equation $\frac{\partial^2 z}{\partial x^2}=x^2\frac{\partial^2 z}{\partial y^2}$ to canonical form.

15. (a) If $\rho>0$ and $\psi(r)=\int_v \frac{\rho(r^1)d\tau^1}{|r-r^1|}$ prove that $\lim_{r=\infty} r\psi(r)=\int_v \rho(r^1)d\tau^1$.

Or

(b) Explain longitudinal vibrations in a bar.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Solve $(1-x^2)y'' - 2xy' + \alpha(\alpha+1)y = 0$, α is a constant.

17. Derive Bessel function of zero order of the first kind.

18. Find the solution of the equation $z = \frac{1}{2}(p^2+q^2) + (p-x)(q-y)$ which passes through x-axis.

19. Explain method of separation of variables, hence solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{1}{k}\frac{\partial z}{\partial t}$.

20. Find the potential function $\psi(x,y,z)$ in the region $0 \le x \le a$, $0 \le y \le b$, $0 \le z \le c$ satisfying the conditions

    (a) $\psi = 0$ on $x = 0, x = a, y = 0, y = b, z = 0$.

    (b) $\psi = f(x,y)$ on $z = c, 0 \le x \le a, 0 \le y \le b$.

AFF–997

Sub. Code
4MMA1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

Elective – MECHANICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write down the D' Alembert's principle.

2. Explain At wood machine.

3. Discuss dissipative function.

4. Write down conservation theorem on total energy.

5. State Hamilton's principle.

6. Prove that the shortest distance between two points in space is straight line.

7. State Bertrand's theorem.

8. Define monogenic system.

9. Write down Hepler's equation.

10. State Kepler's first law.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Derive Lagrange's equation for holonomic system.

Or

(b) Derive Lagrange's equation for the simple pendulum.

12. (a) State and prove conservation theorem for angular momentum.

Or

(b) Show that the two body central force problem can always be reduced to one body problem.

13. (a) State and prove virial theorem.

Or

(b) State and prove Kepler's second law.

14. (a) Derive Lagrange's equation using dissipative function.

Or

(b) Derive Lagrange's equation for a simple particle in space.

15. (a) Derive Kepler's problem.

Or

(b) State and prove Laplace – Runge Lenz vector.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Show that in the absence of total external forces, the total linear momentum of a system is conserved.

17. Discuss about first integrals and solutions of equivalent one body problem.

18. Derive the differential eqn. for the central orbit.

19. Show that in the central force, the orbit is conic.

20. Show that the square of period of revolution of all planets are proportional to cube of semi major axis of the orbits.

AFF–998

Sub. Code
4MMA1E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Mathematics

Elective – PROGRAMMING IN C++

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. State any four principles of OOP.

2. Distinguish between do and while loop.

3. What are objects?

4. How do you declare static data members?

5. What are destructors?

6. What do you mean by overloading?

7. What is redundancy? How to avoid it?

8. Define abstract class.

9. What is a stream?

10. Illustrate an exception.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) List the operators which are not available in "C".

Or

(b) Explain different forms of for loop.

12. (a) Explain function overloading.

Or

(b) Write a program to sort the array of number using bubble sort.

13. (a) Explain different types of constructors.

Or

(b) List all the rules for friend function.

14. (a) Illustrate single inheritance with an example program.

Or

(b) Briefly explain nesting of classes.

15. (a) How do you format an output? Explain.

Or

(b) Explain the statements (i) throw (ii) catch.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain C++ tokens in detail.

17. What is a friend function? Explain.

18. Write a program to add two complex numbers by overloading a binary operator.

19. Explain (a) Multiple inheritance (b) Hierarchical inheritance.

20. List and explain functions for file I/O operations.

AFF–999

Sub. Code
4MMA2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

ALGEBRA — II

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define homomorphism between two vector spaces.

2. Define internal direct sum.

3. Define annihilator of a subspace.

4. Define an inner product space. Provide an example.

5. Define an algebraic number.

6. State Remainder theorem.

7. Define fixed field of G.

8. What is $G(K, F)$?

9. Define a linear transformation.

10. Define right-invertible element.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Prove that $L(S)$ is a subspace of V.

Or

(b) If $v_1, v_2, ..., v_n \in V$ are linearly independent, then prove that every element in their linear span has a unique representation in the form $\lambda_1 v_1 + \lambda_2 v_2 + ... + \lambda_n v_n$ with the $\lambda_i \in F$.

12. (a) If V is finite-dimensional and $V \neq 0 \in V$, then prove that there is an element $f \in \hat{V}$ such that $f(V) \neq 0$.

Or

(b) State and prove Schwarz inequality.

13. (a) Prove that a polynomial of degree n over a field can have at most n roots in any extension field.

Or

(b) Let $f(n) \in F(n)$ be of degree $n \geq 1$. Then prove that there is an extension E of F of degree at most $n!$ in which $f(n)$ has n roots.

14. (a) Prove that the fixed field of G is a subfield of K.

Or

(b) Prove that $G(K, F)$ is a subgroup of the group of all automorphisms of K.

15. (a) If $(vT, vT) = (v, v)$ for all $v \in V$ then prove that T is unitary.

Or

(b) If $T \in A(V)$ has all its characteristic roots in F, then prove that there is a basis of V in which the matrix of T is triangular.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If $v_1, v_2 ..., v_n$ is a basis of V over F and if $w_1, w_2, ...w_n$ in V are linearly independent over F, then prove that $m \leq n$.

17. Let V be a finite-dimensional inner product space; then prove that V has an orthonormal set as F a basis.

18. Prove that the polynomial $f(n) \in F[n]$ has a multiple not if and only if $f(x)$ and $f'(x)$ have a nontrivial common factor.

19. If K is a finite extension of F, then prove that $G(K, F)$ is a finite group and its order; $O(G(K, F))$ satisfies $O(G(K, F)) \leq [K : F]$.

20. If V is n-dimensional over F and if $T \in A(V)$ has all its characteristic roots in F, then prove that T satisfies a polynomial of degree n over F.

AFF–1000

Sub. Code
4MMA2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

ANALYSIS – II

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define unit step function.
2. State the fundamental theorem of calculus.
3. If $\{f_n\}$ and $\{g_n\}$ converge uniformly on a set E, prove that $\{f_n + g_n\}$ converges uniformly on E.
4. Define equicontinuous function.
5. State Stirling's formula.
6. Find the limit $\lim_{x \to 0} \frac{x - \sin x}{\tan x - x}$.
7. If $m^* E = 0$ then prove that E is measurable.
8. Prove that the product of two measurable extended real-valued function is measurable.
9. State bounded convergence theorem.
10. Show that if f is integrable over E then $|f|$ is also integrable.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Prove that $f \in \mathrm{R}(\alpha)$ on $[a,b]$ if and only if for every $\in > 0$ there exists a partition P such that $U(p,f,\alpha) - L(p,f,\alpha) < \in$.

Or

(b) If $f \in R(\alpha)$ and $g \in R(\alpha)$ on $[a,b]$. Then prove that :

(i) $f\,g \in R(\alpha)$

(ii) $|f| \in R(\alpha)$ and

(iii) $\left|\int_a^b f d\alpha\right| \le \int_a^b |f| d\alpha$.

12. (a) The sequence of function $\{f_n\}$ defined on E. Prove that $\{f_n\}$ converges uniformly on E if and only if for every $\in > 0$ there exists an integer N such that $m \ge N$, $n \ge N$, $x \in E$ implies $|f_n(x) - f_n(x)| \le \in$.

Or

(b) If K is a compact metric space, if $f_n \mathcal{C}$(K) for $n = 1,2,3...$ and if $\{f_n\}$ converges uniformly on K, then prove that $\{f_n\}$ is equicontinuous on K.

13. (a) Let $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = E(x)$ be defined on R'. Then prove that :

(i) e^x is continuous and differentiable for all x

(ii) $(e^x)^1 = e^x$

(iii) e^x is a strictly increasing function of x and $e^x > 0$

Or

(b) If f is a positive function on $(0, \infty)$ such that

(i) $f(x+1) = x\,f(x)$

(ii) $f(1) = 1$

(iii) $\log f$ is convex then prove that $f(x) = \Gamma(x)$.

14. (a) Let $< Ei >$ be a sequence of measurable sets. Then prove that $\mathrm{m(UEi)} \le \sum \mathrm{mEi}$. If the sets E_n are pair wise disjoint, then $\mathrm{m(UEi)} = \sum \mathrm{mEi}$.

Or

(b) If E_1 and E_2 are measurable then prove that $\mathrm{E_1 U E_2}$ is also measurable.

15. (a) If f and g are bounded measurable functions defined on a set E of finite measure then prove that

(i) $\int_E (af + bg) = a\int_E f + b\int_E g$

(ii) If $f = g$ a.e. then $\int_E f = \int_E g$.

Or

(b) State and prove monotone convergence theorem.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If r' is continuous on $[a,b]$ then prove that γ is rectifiable and $\wedge(\gamma) = \int_a^b |\gamma(t)|\, dt$.

17. State and prove Stone-Weierstrass theorem.

18. State and prove Parsevals theorem.

19. Prove that the outer measure of an interval is its length.

20. Let f and g be integrable over E. Then prove that

    (a) The function cf is integrable over E and $\int_E cf = C\int_E f$.

    (b) The function $f+g$ is integrable over E and $\int_E f+g = \int_E f + \int_E g$

    (c) If $f \le g$ a.e, then $\int_E f \le \int_E g$

    (d) If A and B are disjoint measurable sets contained in E, then $\int_{AUB} f = \int_A f + \int_B f$.

AFF–1001

Sub. Code
4MMA2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

PROBABILITY AND STATISTICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Let $f(x)=1$, $0<x<1$, zero else where, be the p.d.f of X. Find distribution function.

2. Let X have the p.d.f $f(x)=\frac{x+1}{2}-1<x<1$, zero else where find variance of x.

3. Write down the pdf of Binomial distribution.

4. Define Bivariate Normal distribution.

5. Let $f(x_1, x_2)=4x_1, x_2, 0<x_1<1, 0<x_2<1$, zero else where be pdf of X_1 and X_2 find $\Pr\left(0<X_1,<\frac{1}{2}, \frac{1}{4}<X_2<1\right)$.

6. Let $f(x_1, x_2)=6x_2$, $0<x_2<x_1<1$, zero else where find marginal p.d.f of X_1. The conditional pdf of X_2 given $X_1=x_1$.

7. Define t distribution.

8. Define f distribution.

9. State central limit theorem.

10. Let Zn be $\chi^2(n)$ let $Wn = \frac{Z_n}{n^2}$. Find limiting distribution of W_n.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Let $f(x) = \frac{3!}{x!(3-x)!}$ $x = 0, 1, 2, 3$ and $f(x) = 0$ e.w. Find mean and variance.

Or

(b) Let x have p.d.f. $f(x) = 2(1-x)$ $0 < x < 1$, zero else where, Find $E(X)$, $E(X^2)$ and $E(6X + 3X^2)$.

12. (a) Show that the random variables X_1 and X_2 with $f(x_1, x_2) = 12x_1x_2(1-x_2)$ $0 < x_1 < 1$ $0 < x_2 < 1$, zero else where are independent.

Or

(b) Let the random variables X_1 and X_2 have a joint pdf $f(x_1, x_2) = x_1 + x_2$ $0 < x_1 < 1$ $0 < x_2 < 1$ zero else where compute correlation coefficient.

13. (a) If X is $b(n, p)$ show that $E\left(\frac{x}{n}\right) = p$,

$$E\left(\frac{x}{n} - p\right)^2 = \frac{p(1-p)}{n}.$$

Or

(b) Derive the m.g.f of Normal distribution.

14. (a) Show that $s^2 = \frac{1}{n}\Sigma(x_i - \bar{x})^2 = \frac{1}{n}\Sigma x_i^2 - \bar{x}^2$ where $\bar{x} = \frac{\Sigma x_i}{n}$.

Or

(b) Let X have the pdf $f(x) = \left(\frac{1}{2}\right)^x$ $x = 1, 2, ...$ find the pdf of $Y = X^3$.

15. (a) Let $\overline{X}_n$ denote mean of random sample of size n from $N(\mu, \sigma^2)$ find limiting distribution of $\overline{X}_n$.

Or

(b) Let $F_n(u)$ denote distribution sum of random variable U_n. Let U_n coverages in probability to the constant C to show that $\frac{Un}{C}$ converges in probability of 1.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. State and prove Chehyshev's inequality.

17. Let X_1 and X_2 denote r.v. that have joint pdf $f(x_1, x_2)$ and marginal p.d.f $f_1(x_1)$ and $f_2(x_2)$ respectively $\wedge$ show that x_1 and x_2 are independent iff $m(t_1, t_2) = m(t_1, 0)$. $m(0, t_2)$.

18. Define the pdf of Normal distribution.

19. Derive the pdf of F distributions.

20. State and prove central limit theorem.

AFF–1003

Sub. Code
4MMA2E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

Elective : GRAPH THEORY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Show that if G is simple, then $\varepsilon \leq \binom{\gamma}{2}$.

2. Prove that any two longest paths have a common vertex in a connected graph.

3. Define edge connectivity. Give an example.

4. Draw a graph which is Eulerian but not Hamiltonian.

5. Find a maximum matching in $K_{3,3}$.

6. If G is bipartite prove that $x' = \Delta$.

7. Define $r(k,l)$.

8. Show that for any graph G, $\chi \leq \Delta + 1$.

9. Define G^*.

10. Prove that K_5 is non planar.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Let G be simple. Show that $\varepsilon = \binom{\gamma}{2}$ iff G is complete.

Or

(b) If e is a link of G, prove that $\tau(G) = \tau(G - e) + \tau(G.e)$.

12. (a) Prove that $k \le k' \le \delta$.

Or

(b) Show that a nonempty connected graph is Eulerian iff it has no vertices of odd degree.

13. (a) State and prove Hall's theorem.

Or

(b) Find the number of different perfect matchings in K_{2n}.

14. (a) If $\delta > 0$, prove that $\alpha' + \beta' = \gamma$.

Or

(b) Prove that in a critical graph no vertex cut is a clique.

15. (a) State and prove Euler's formula.

Or

(b) Prove $K_{3,3}$ is non-planar (by construction).

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. State and prove characterisation theorem of bipartite graphs.

17. State and prove Chvatal's theorem.

18. State and prove Vizing's theorem.

19. Prove that if G is a connected simple graph and is neither an odd cycle non a complete graph, then $\chi \leq \Delta$.

20. Prove that every planar graph is 5-vertex colourable.

AFF–1005

Sub. Code
4MMA2E4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Mathematics

Elective – DISCRETE MATHEMATICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Give an example of semigroup but not a monoid.

2. Define submonoid. Give an example.

3. Distinguish between atomic and compound statements.

4. List the uses of normal forms.

5. What are quantifiers?

6. State EG and US rules.

7. Draw the Harse diagram of a lattice.

8. Give an example of a modular lattice.

9. Give an example for a Boolean algebra.

10. Draw NAND, NOR gates.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) How many commutative binary operations can be defined on a set A with 'n' elements.

Or

(b) Show that if g is a homomorphism from a commutative semigroup $(S, *)$ on to a semigroup (T, Δ) then (T, Δ) is also commutative.

12. (a) Show that $(\neg P \wedge (Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$ without using truth table.

Or

(b) Prove that $\{\uparrow\}, \{\downarrow\}$ are functionally complete.

13. (a) Show that S is a valid inference from $P \rightarrow \neg Q, QVR, \neg S \rightarrow P$ and $\neg R$.

Or

(b) Symbolise "every one who is healthy can do all kinds of work".

14. (a) Prove that every chain is a lattice.

Or

(b) Show that every distributive lattice is modular.

15. (a) Prove that it x, y are elements in a Boolean algebra, $x \leq y \Leftrightarrow x' \geq y'$.

Or

(b) Express $P(x_1, x_2, x_3) = x_1 \vee x_2$ in an equivalent sum-of-products canonical form.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Show that monoid homomorphism preserves the property of invertibility.

    (b) Show that for any commutative monoid (M,*) the set of idempotent elements of M forms a submonoid.

17. Obtain PDNF of $P \rightarrow ((P \leftrightarrow Q) \wedge \neg(\neg Q \vee \neg P))$.

18. Show that if $P \rightarrow q, q \rightarrow r, \neg(P \wedge r)$ and $P \vee r$ then r.

19. Prove that direct product of any two distributive lattices is a distributive lattice.

20. Simplify $x_1x_3 + x_1'x_3x_4 + x_2x_3'x_4 + x_2'x_3x_4$ using karnaugh map.

AFF–1007

Sub. Code
4MMA3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

COMPLEX ANALYSIS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define analytic function. Give an example.
2. What is parallel translation?
3. Evaluate $\int_{\gamma} x\,dz$, γ is a line segment from 0 to 1+i.
4. Find $\int_{|z|=1}^{\gamma} \frac{e^z}{z} dz$.
5. Define isolated singularity with an example.
6. Define mesomorphic function, give and example.
7. Find the poles of $\frac{1}{z^2+5z+6}$.
8. Define residue of $f(z)$ at an isolated singularity.
9. Write the Taylor's series expansion of arc tan Z.
10. Define genesis.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Prove that the functions $f(z)$ and $\overline{f(\bar{z})}$ are simultaneously analytic.

Or

(b) State and prove Lucas theorem.

12. (a) Show that $\left|\int_{\gamma} f dz\right| \leq \int_{\gamma} |f| \, |dz|$.

Or

(b) State and prove Morera's theorem.

13. (a) State and prove Weierstrass theorem.

Or

(b) State and prove Maximum principle.

14. (a) State and prove Residue theorem.

Or

(b) Find the residues of $\frac{1}{\sin^2 z}$.

15. (a) Prove that Laurent development is unique.

Or

(b) State and prove necessary and sufficient condition for the absolute convergence of $\prod_{1}^{\infty}(1 + a_n)$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Prove that the cross ratio (z_1, z_2, z_3, z_4) is real iff the four points lie on a circle or on a straight line.

17. State and prove Cauchy's theorem for a rectangle.

18. State and prove Weierstrass theorem on essential singularity.

19. Prove that $\int_0^{\pi} \log \sin x \, dx = -\pi \log 2$.

20. State and prove Jensen's formula.

AFF–1008

Sub. Code
4MMA3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

TOPOLOGY — I

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define the standard topology on the real line.

2. Show that the projection maps an open maps.

3. Define linear continuum.

4. Give an example of a curve which is connected but not path connected.

5. State the maximum and minimum value theorem.

6. Define sequentially compact.

7. Define the Lindelof space. Give an example.

8. Discuss the necessity of closedness in Tietze theorem.

9. Discuss the second countability axiom on a topological space.

10. Explain the concept of topological imbedding.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Let $\mathcal{B}$ be a basis for a topology τ on a set X. Then prove that τ equals the collection of all unions of element of $\mathcal{B}$.

Or

(b) State and prove the necessary and sufficient condition for a subset of a topological space to be closed.

12. (a) State and prove the pasting Lemma.

Or

(b) State and prove the sequence lemma.

13. (a) State and prove the intermediate value theorem.

Or

(b) Prove that every compact subspace of a Hausdorff space is closed.

14. (a) State and prove the uniform continuity theorem.

Or

(b) Prove that the compactness implies limit point compactness.

15. (a) Suppose if X has a countable basis them prove that there exists a countable subset of X that is dense in X.

Or

(b) Prove that every well-ordered in set X is normal in the order topology.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Prove that the topologies on $\mathbb{R}^n$ induced by the Euclidean metric d and square metric ρ are the same as the product topology on $\mathbb{R}^n$.

17. (a) Let A be a subset of the topological space X. Then prove that $x \in \overline{A}$ if and only if every open set U containing x intersects A. (5)

    (b) Prove that the collection of subsets of X whose union equals X forms a topology on X. (5)

18. Prove that the real line $\mathbb{R}$ is connected.

19. Prove that the product of finitely many compact spases is compact.

20. State and prove Uryshon's lemma.

AFF–1009

Sub. Code
4MMA3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

OPERATIONS RESEARCH

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. What is three-jug puzzle?

2. What is a critical activity?

3. Define purchasing cost.

4. Define holding cost.

5. Define an exponential distribution.

6. What is lack of memory property?

7. Find p_0 of the model $(M/M/C):(GD/N/\infty)$; $C \le N$.

8. Find L_s of the model $(M/M/1):(GD/N/\infty)$.

9. What are the solution methods of non-linear programming?

10. What is the use of gradient algorithm?

Section B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain Dijkstra's algorithm.

Or

(b) Explain Floyd's algorithm.

12. (a) Neon lights on the U of A campus are replaced of the rate of 100 units per day. The physical plant orders the neon lights periodically. It costs \$ 100 to initiate a purchase order. A neon light kept in storage is estimated to cost about \$0.02 per day. The lead time between placing and receiving on order is 12 days. Determine the optimal inventory policy for ordering the neon lights.

Or

(b) Explain silver-meal heuristic algorithm.

13. (a) Describe pure-death model.

Or

(b) If one arrival occurs every 10 minutes, determine the average arrival rate per hour λ, and the average interarrival time in hours.

14. (a) Describe the standard queue notation $(a/b/c):(d/e/f)$.

Or

(b) Describe the queuing model $(M/M/1):(GD/\infty/\infty)$.

15. (a) Explain the separable convex programming method.

Or

(b) Use separable programming algorithm to the non-linear programming problem

Maximize $z = x_1 + x_2^2$

Subject to

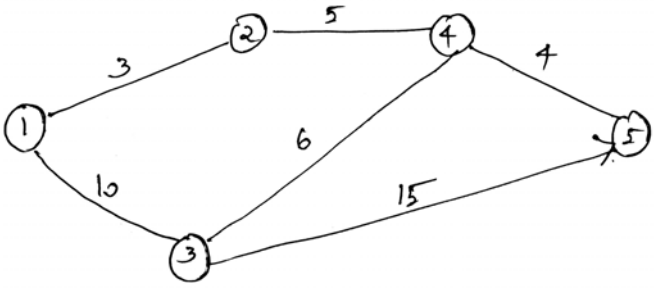
$3x_1^2 + 2x_2^2 \leq 9,$

$x_1 \geq 0,\ x_2 \geq 0.$

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Find the shortest routes between every two nodes of the following network using Floyd's algorithm. The distances (in miles) are given on the arcs. Arc (3, 5) is directional, so that no traffic is allowed firm node 5 to node 3. All the other arcs allow two-way traffic.



17. The following table provides the data for a 3-period inventory situation.

Period i	Demand D_i (units)	Setup cost K_i \$	Holding cost h_i (\$)
1	3	3	1
2	2	7	3
3	4	6	2

The demand occurs in discrete units and the starting inventory is $x_1 = 1$ unit. The unit production cost is \$ 10 for the first 3 units and \$ 20 for each additional unit. Determine the optimal inventory policy.

18. Babies are born in a sparsely populated state at the rate of one birth every 12 minutes. The time between births follows an exponential distribution. Find the following

    (a) The average number of births per year

    (b) The probability that no births will occur in any one day

    (c) The probability of issuing 50 birth certificates in 3 hours given that 40 certificates were issued during the first 2 hours of the 3-hour period.

19. A barber shop has two barbers and three chairs for customers. Assume that customers arrive in a Poisson fashion of a rate of 5 per hour and that each barber services customers according to an exponential distribution with mean of 15 minutes. Further if a customer arrives and if there are no empty chairs in the shop, he will leave. What is the probability that the shop is empty? What is the expected number of customers in the shop?

20. Solve :

    Maximize $z = x_1 + x_2^4$

    Subject to

    $3x_1 + 2x_2^2 \leq 9$

    $x_1, x_2 \geq 0.$

AFF–1010

Sub. Code
4MMA3C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

NUMBER THEORY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Prove that if a prime p does not divide a then $(p,a)=1$.

2. If $(a,b)=(a,c)=1$, prove that $(a,b,c)=1$.

3. Define the Mobius function.

4. For all f, prove that $I*f=f*I=f$ where I is the identity function.

5. Prove that for $x \geq 1$, $\sum_{n \leq x} \mu(n)\left[\frac{x}{n}\right]=1$.

6. If $0<y<1$, what are the possible values of $[x]-[x-y]$?

7. Show that if $ac \equiv bc (\bmod m)$ and if $d=(m,c)$ then $a \equiv b (\bmod \frac{m}{d})$.

8. Show the congruence $5x \equiv 3 (\bmod 24)$.

9. Find the value of $\left(\frac{191}{3}\right)$.

10. Determine whether –104 is a quadratic residue or non-residue of the prime 997.

Part B $(5 \times 5 = 25)$

Answer **all** questions

11. (a) State and Prove fundamental theorem of arithmetic.

Or

(b) Given any two integers a and b, there is a common divisor d of a and b of the form $d = ax + by$ where x and y are integers. Moreover every common divisor of a and b divides this d.

12. (a) If $n \geq 1$, prove that $\sum_{d/n} \phi(d) = n$.

Or

(b) State and Prove Selberg Identify.

13. (a) State and Prove Euler's summation formula.

Or

(b) For $x \geq 2$, show that $\sum_{p \leq x} [\frac{x}{p}] \log p = x \log x + O(x)$.

14. (a) State and Prove Chinese Remainder Theorem.

Or

(b) If a prime does not divide a, then prove that $a^{p-1} \equiv 1 (\text{mod } p)$.

15. (a) Let p be an odd prime. Prove that for all n, $(n/p) = n^{(p-1)/2} (\text{mod } p)$.

Or

(b) State and Prove quadratic reciprocity law for Jacobi Symbols.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that the infinite series $\sum_{n=1}^{\infty} \frac{1}{p^n}$ diverges.

17. If f is an arithmetical function with $f(1) \neq 0$, there is a unique arithmetical function f^{-1}, called the Dirichlet inverse of f, such that $f * f^{-1} = f^{-1} * f = I$. Moreover, f^{-1} is given by the recursion formula $f^{-1}(1) = \frac{1}{f(1)}$,

$$f^{-1}(n) = \frac{-1}{f(1)} \sum_{\substack{d/n \\ d<n}} f\left(\frac{n}{d}\right) f^{-1}(d), for\, n > 1.$$

18. For all $x \geq 1$, prove that $\sum_{n \leq x} d(n) = x \log x + (2c-1)x + O(\sqrt{x})$ where C is the Euler's constant.

19. Assume $(a,m) = d$ and suppose that d/b. Then prove that the linear congruence $ax \equiv b (\bmod m)$ has exactly d solutions modulo m. These are given by $t, t + \frac{m}{d}, t + 2\frac{m}{d}, \ldots\ldots, t + (d-1)\frac{m}{d}$; where t is the solution, unique modulo $\frac{m}{d}$, of the congruence

$$\frac{a}{d} x \equiv \frac{b}{d} \left(\bmod \frac{m}{d} \right).$$

20. Prove that the Diophantine equation $y^2 = x^3 + k$ has no solutions if k has the form $k = (4n-1)^3 - 4m^2$, where m and n are integers such that no prime $p \equiv -1 (\bmod 4)$ divides m.

AFF–1011

Sub. Code
4MMA3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

COMBINATORIAL MATHEMATICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer the following.

1. Discuss about the generating functions and indicators functions of a sequence.

2. Find the number of r-permutations of n distinct objects with unlimited repetitions using exponential enumerator.

3. Define a recurrence relation.

4. Give an example of a recurrence relations with two indices.

5. Discuss the derangement of the integers.

6. State the principle of inclusion and exclusion.

7. Define an equivalence relation.

8. Define the weight of a pattern.

9. Define a complete balanced block design.

10. Define isomorphism of block designs.

Part B $(5 \times 5 = 25)$

Answer the following.

11. (a) What is the coefficient of x^{23} in $\left(1+x^5+x^9\right)^{100}$?

Or

(b) Find the number of r-digit quaternary sequences in which each of the digits 1,2 and 3 appears atleast once.

12. (a) Solve the recurrence relation for the Fibonacci sequence.

Or

(b) Find the number of n-digit binary sequences that have the pattern 010 occurring at the nth digit.

13. (a) Find the number of n-digit ternary sequences that have an even number of 0's.

Or

(b) In how many ways can the integers 1,2,3,4,5,6,7,8 and 9 be permutated such that no odd integer will be in its natural position?

14. (a) Prove that the binary operation of composition of permutation form a group.

Or

(b) Find the no. of ways of painting the four faces of the pyramid with two colors of paints.

15. (a) Discuss the context of perfect difference set and perfect difference family with examples.

Or

(b) State and prove Fisher's inequality.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Show that the ordinary generating function of the sequence $\binom{0}{0},\binom{2}{1},\binom{4}{2},\binom{6}{3},\ldots\binom{2r}{r}$ is $(1-4x)^{-1/2}$.

17. Solve the recurrence relation $a_n = a_{n-1} + 2(n-1)$ where $a_0 = a_1 = 2$ and $n \geq 1$.

18. Find the number of integers between 1 and 250 that are not divisible by 2,3,5 and 7.

19. State and prove Burnside theorem.

20. Prove that for every non-empty balanced incomplete block design $\lambda < r$.

AFF–1012

Sub. Code
4MMA3E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

STOCHASTIC PROCESS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define stationary process.

2. Define sub-martingale.

3. State the strong Markov property.

4. Define egodic Markov chain.

5. Explain stationary distribution.

6. What is the distribution of the interval between two successive occurrences of Poisson process $\{N(t), t > 0\}$ having parameter 'X'?

7. What is Poisson cluster process?

8. When a Birth and Death process is known as immigration – emigration process?

9. Define stopping time.

10. Write Wald's equation.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain Markov process, Markov chain.

Or

(b) Show that the process $X(t) = A\cos wt + B\sin wt$ where A, B are uncorrelated r.v's each with mean '0' and variance '1'. w is constant, is covariance stationary.

12. (a) Show that the state j is persistent or transient according as $\sum P_{ij}^{(n)} = \infty w < \infty$.

Or

(b) If state j is persistent prove that for every state k that can be reached from state 'j', $F_{Kj} = 1$.

13. (a) List the postulates for Poisson process.

Or

(b) Prove that sum of two independent Poisson processes is a Poisson process.

14. (a) Explain Yule-Furry process.

Or

(b) Prove that the p.g.f of a non-homogeneous process $\{N(t), t \geq 0\}$ is $Q(s,t) = \exp\{m(t)(s-1)\}$.

15. (a) Show that the renewal function M satisfies $M(t) = F(t) + \int_0^t M(t-x)\, dF(x)$.

Or

(b) State and prove central limit theorem for renewals.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Define Gaussian process. Hence show that it is strictly stationary if it is stationary.

17. Explain the method of determining higher transition probabilities.

18. State and prove Ergodic theorem.

19. Explain Birth and Death process.

20. State and prove elementary renewal theorem.

AFF–1013

Sub. Code
4MMA3E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Mathematics

FUZZY MATHEMATICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define a characteristic function.

2. State distributive laws of crisp sets.

3. Define complement of a fuzzy set.

4. What is a sugeno class?

5. Define Cartesian product of two crisp sets with an example.

6. Define a fuzzy binary relation. Give an example.

7. State any two axioms of fuzzy measure.

8. Define a plausibility measure.

9. What is symmetric fuzzy linear programming provide an example?

10. What is fuzzy dynamic programming? Give the basic structure of the model.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Define a fuzzy set. Give an example.

Or

(b) Explain about the fuzzy measure with an example.

12. (a) Prove that every fuzzy complement has at most one equilibrium.

Or

(b) If C is a continuous fuzzy complement, then prove that C has a unique equilibrium.

13. (a) Write short notes on fuzzy relation equations.

Or

(b) Write short notes on Hasse diagrams.

14. (a) Prove that a belief measure Bel on a finite power set $P(X)$ is a probability measure if and only if its basic assignments m is given by $m(\{x\}) = Bel(\{x\})$ and $M(A) = 0$ for all subsets of X that are not singletons

Or

(b) Prove that every possibility measure π on $P(X)$ can be uniquely determined by a possibility distribution function

$r : X \to [0, 1]$ via the formula

$\pi(A) = \underset{x \in A}{man}\, r(x)$ for each $A \in P(X)$.

15. (a) Write short notes on multi objective decision making.

Or

(b) Write short notes on Multi Attribute Decision Making (MADM).

Part C (3 × 10 = 30)

Answer any **three** questions.

16. What is a Boolean algebra? Write any five properties of Boolean algebra.

17. Prove that fuzzy set operations of union intersection, and continuous and complement that satisfy the law of excluded middle and the law of contradiction are not idempotent or distributive.

18. Explain about ordinary fuzzy homomorphism and fuzzy homomorphisms.

19. Discuss the relationship among classes of fuzzy measures.

20. Discuss about symmetric fuzzy LP with necessary examples.